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Appendix to the CV: Selected Methods

Summaries of four methods and one project in progress, developed between April and November 2026 for a two-arm clinical TMS study. None is published, and results are exploratory unless stated. Full reports: github.com/sowrder/Formulation-landscape.

How this work is produced *method and attribution*

Method. The designs, definitions and decisions are the author's. Code is written with AI assistance to those designs. Results are verified through diagnostics and mathematical checks the author specifies (exact round trips, known cases re-solved, held-out calibration, rotation and permutation nulls) and not by reading code line by line. The longer reports mark each statement by source: decided by the author, read from data, derived, or inferred.

Rationale. Systems of this scope could not be written by hand by one person in the time available. Delegating the writing while keeping the design and the checks makes the scope possible, and it places the risk where it can be measured: in whether outputs pass tests with known answers. The AI contributes established methods; how they are combined, and what counts as correct, is set by the author.

1. Data-driven segmentation of gaze into regions of interest *eye tracking*

Problem. Predefined areas of interest impose the analyst's labels on where people look. To study face viewing in free-viewing images, regions are derived from the gaze data alone and only afterwards compared with the image content.

Method. For each participant and image, gaze samples (500 Hz, with saccades and missing data removed) give a two-dimensional Gaussian kernel density estimate p. A region centre μ_k is a local maximum of p above a density floor at which the Hessian of $-\log p$ is positive definite. The construction comes from a diffusion with zero probability flux, where drift $f = D\,\nabla \log p$, so each maximum is a local quadratic well $\Lambda_k = -D\,\nabla\nabla^T \log p$ at μ_k. A sample x belongs to the nearest well if $\tfrac{1}{2}\,(x-\mu_k)^T\,[-\nabla\nabla^T \log p]\,(x-\mu_k) \le 1$. The diffusion constant D cancels, so the partition depends on the density alone; the approach is related to mean-shift region methods (Santella and DeCarlo, 2004). Consecutive samples in one region form a visit, with a dwell time and a re-entry count.

Validation. Face outlines were fitted from facial landmarks using the images only. Region centres fell inside faces 2.9 times as often as the same participant's regions taken from a different image (95% CI 2.59–3.28). Partitions agreed across three kernel widths (adjusted Rand index 0.74–0.83).

Results. In 29 sessions from 18 participants, with each effect fitted per session and tested across sessions: faces held gaze 23% longer than other regions (27 of 29 sessions); dwell fell with distance from the face centre (22 of 29); face size predicted which faces were looked at in every session; and sustained looking at a face declined after repeated returns (25 of 29), with no such change off faces. The face and face-size effects agree with published findings on face viewing, which supports the segmentation.

Limits. The dynamical reading fails: the curvature of a region does not predict dwell time (log–log slope −0.13 against a predicted −1), so the model is a segmentation tool, not a model of gaze dynamics. Sessions from the same participant are treated as independent; a participant-level analysis is pending. Secondary effects are uncorrected for multiple tests.

Materials. Full report (6 pp.): github.com/sowrder/Formulation-landscape/gaze_segmentation.

2. Decomposing a longitudinal export by visit schedule *REDCap*

Problem. A longitudinal REDCap export is one wide table in which each instrument is collected at its own subset of visits, so most cells are empty by design and blanks cannot be told apart from missing data.

Formulation. Each instrument i has an ordered event sequence E(i) and a set of fields; fields are disjoint across instruments. Define $i \sim j$ when $E(i) = E(j)$. This is an equivalence relation, and its classes are called blocks. Blocks are in one-to-one correspondence with the distinct event sequences, and a block's fields are the union of its instruments' fields. The export then factors into one dense participants × events × fields array per block, with a field-to-block index for lookup. Inside a block every blank cell is missing data.

Use. The structure feeds daily, de-identified tables for 27 questionnaire measures (one row per time point, one column per participant) in two cohorts, with a third being added. It applies to any export keyed by subject and time point in which variables follow schedules.

Limits. This regroups information that REDCap's instrument–event mapping already provides; the contribution is the block structure and its index, not a new result. Instruments that share a schedule are deliberately not distinguished within a block.

Materials. Formulation notes (3 pp.): github.com/sowrder/Formulation-landscape/redcap_blocks.

3. Scripted control of GraphPad Prism graphs *file-format analysis*

Problem. Figures for many measures must share one look and be refreshed with every export. Prism stores tables as readable JSON and CSV, but graph styling sits in a binary component for which no specification was found.

Method. Controlled experiment: one change made in Prism per saved file, each file compared with the previous one, and each candidate field then confirmed by writing it by script and rendering the result. Every write first asserts the current value and locates its field by structure.

Findings. A data column binds to a graph series through an identifier, not by position. Each series has a fixed-size style record holding symbol, size, line pattern, line type and thickness. Colours live in a compressed tagged tree and are addressed by key name; parsing and rebuilding that tree reproduced every tested file byte for byte.

Result. Data refresh never touches the binary component. A style specification produced in a small designer interface restyles prepared templates by script; graphs with up to 217 series rendered correctly.

Limits. New series cannot yet be created by script and require one save in Prism. Some field positions differ between file families, and a few fields remain inferred pending a confirming render.

Materials. Format reference with rendered test figures (12 pp.): github.com/sowrder/Formulation-landscape/prism_format.

4. A calendar-driven framework for study data, applied to two clinical studies *research operations*

Problem. Each participant produces audio, MRI, EEG, eye-tracking, behavioural, neuronavigation and treatment recordings on different machines. Renaming, de-identifying, uploading and checking them had been manual work.

General core. (1) Every participant, visit and data stream is a node whose state can only move forward, and only the process that owns a node may judge it. (2) The schedule is the authority: the study calendar defines what should exist, and timers keyed by kind, subject and date are armed once. (3) One machine executes; other machines hold read-only views and can only send it signals. (4) Storage locations carry a permission map (some are read-only), and reads consult a cached index before any live listing. (5) Steps that need a person are logged confirmations that never block the automation, and only a person can mark an item done. (6) Work on other machines follows an exit-code contract. (7) Code and state travel between machines versioned and encrypted.

Application. A study is written into the core as its roster and identifier map, its visit diagram (which defines the expected data), its storage locations, its machines, its daily pass times, its human approval points and its drivers for outside services. The two studies it runs have different visit schedules and data streams and share the core: 28 processes on four machines. Its requirements are cloud storage, separately stored behavioural data and an institutional network.

Build history. A first version (about 9,000 lines of Python with a terminal interface) ran by mid-October 2026. From 19 October to 11 November it was hardened in stages: state logic mapped and repaired in a sandbox that replays a simulated week; execution moved to an always-on machine with observer roles; source checks added before any overwrite; intake automated (transcription and stimulus preparation, with clinician approval); read-only protections and health checks for outside services; validation redefined as a logged human step; then scheduled preprocessing on a GPU workstation. Every process was validated live on 5 November, and by 11 November every step of the daily loop completed without error.

Failures found. Two silent failures shaped the design. A calendar read that returned an empty calendar left an invitation sender inactive for about two weeks, and a full temporary disk stopped state synchronisation for two days. Each led to a guard: sources are validated before use, and failed transfers are reported.

Use. The system is in continuous daily use in the project and replaced steps previously done by hand.

Limits. One maintainer and no independent code review. Generality is a design property shown on two studies at one site; it has not been tested on a third study or at another institution. The system does not use a general workflow orchestrator.

Materials. Architecture and build-history reference: github.com/sowrder/Formulation-landscape/study_data_framework. Related public tool: github.com/sowrder/OrganizePile.

5. Work in progress: a spatial dose model of the treatment *first full run; no supported effect yet*

This is the current project. The pipeline has run end to end on one cohort, and its first statistical result is a null, reported as such.

Question. Does the way the stimulation field lands on each person’s own brain relate to how symptoms change across the treatment week?

Data. Twelve participants of one cohort are processed. Treatment is five days of six sessions; each session delivers 3,000 pulses in 75 trains from a closed-loop system that times pulses to the participant’s alpha rhythm, with a resting recording before each session. Coil position was recorded by hand a few times per session: about 28 samples per participant against roughly 2,200 gaps between trains.

Anatomy arm. To give “the same place” a meaning across brains that does not depend on where the coil sat, each brain is divided into 72 regions defined by direction from the centroid of its cortical volume, in an anatomical frame fitted to that

brain, from one template of directions averaged over 29 participants. Majority atlas labels of the regions agreed across participants at a median of 0.97 (0.72 for an earlier version). Under a rotation test, agreement was unchanged up to 2° of frame error and fell to a chance floor of 0.13 at random orientation. The regions are a shared coordinate system and do not identify individuals.

Treatment arm. Recordings split by restarts are rebuilt into one file per session on one clock, with every pulse time-stamped and sample values unchanged. Each pulse train is given a coil position labelled as recorded, carried forward or uncertain. Uncertain stretches are filled from a heavy-tailed population model of placement error using one bank of 50 random draws shared by all participants, so that differences between people do not come from different random numbers. A field simulation is run for every position used.

Contribution. Dose is pulses multiplied by field magnitude, accumulated per day on an equal-solid-angle grid of directions for the cortex and four depth layers. Control variants reweight sessions (pulses alone, stimulator output, closeness to the alpha frequency) while pulses, positions and fields stay fixed, and are compared on identical rows by AICc.

First result. On 48 participant-days from 12 participants the spatial terms did not improve the model for any of five outcomes (smallest permutation $p = 0.07$). A participant's daily dose maps differ by only 3–8%, so the design has little power to follow day-to-day change. The result is read as "not supported yet", not as absence.

Next. A second cohort of 20 participants is entering the same pipeline. Covariate maps are being rerun with compositional shares, participant and day effects removed and family-wise permutation control; alpha-frequency estimates recomputed offline will be added as a further variant.

Open. The mapping from the cached field to stimulator output is assumed, and the unit of the closed-loop error measure is unsettled. Fields cover about 70 mm around the target. The dose model is linear and additive.

Materials. Methods draft (24 pp.) and stage-by-stage scheme (8 pp.): github.com/sowrder/Formulation-landscape/spatial_dose_model.